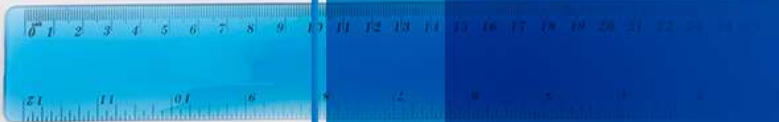


A top-down view of various school supplies arranged on a white surface. In the top left, a black mesh pen holder contains several pens and pencils. Below it, a pair of blue-handled scissors is open. To the right of the scissors is a red pencil, a white eraser, and a small blue sharpener. Further right is a black calculator with a small screen and various function buttons. Next to the calculator is a blue stapler. A long blue ruler is positioned horizontally at the bottom. Above the ruler, there are two pens (one blue, one silver), a pink sticky note, and a blue protractor. A blue set square is visible in the top right corner.

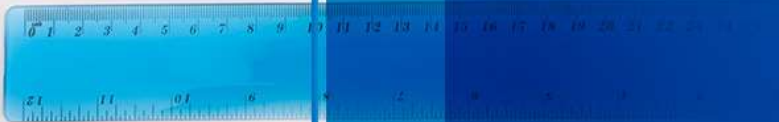
Week-15

Introduction to Database Systems

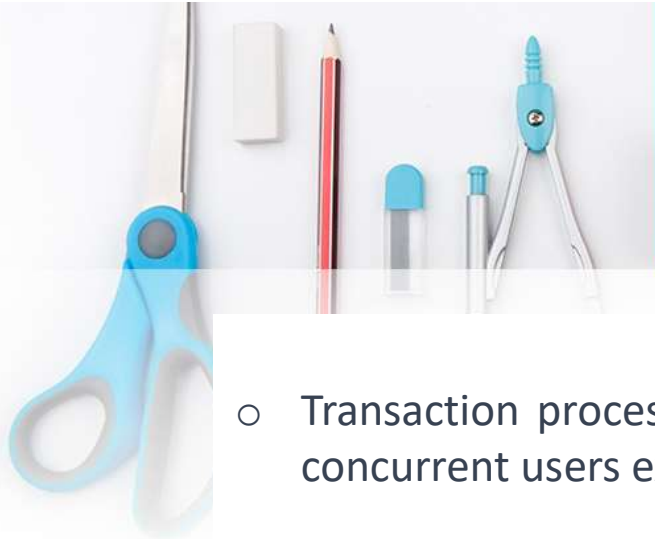
Shah Nawaz



- Introduction to Transaction Processing Concepts
- Concurrency Control Techniques



Introduction to Transaction Processing Concepts



Introduction to Transaction Processing Concepts

- Transaction processing systems are systems with large databases and hundreds of concurrent users executing database transactions.
- **Examples :**
 - **airline reservations,**
 - **banking,**
 - **credit card processing,**
 - **online retail purchasing,**
 - **Stock markets etc**
- These systems require high availability and fast response time for hundreds of concurrent users

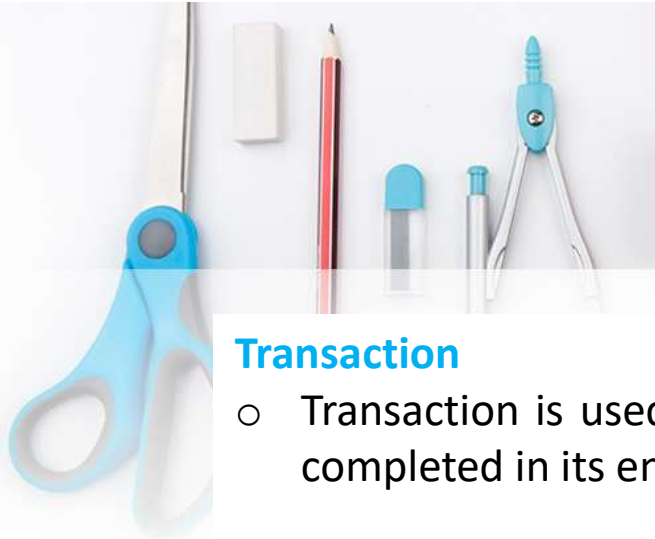


Introduction to Transaction Processing Concepts

DBMS Types

- **Single-user :**
- A DBMS is single-user if at most one user at a time can use the system,
- Single-user DBMSs are mostly restricted to personal computer systems

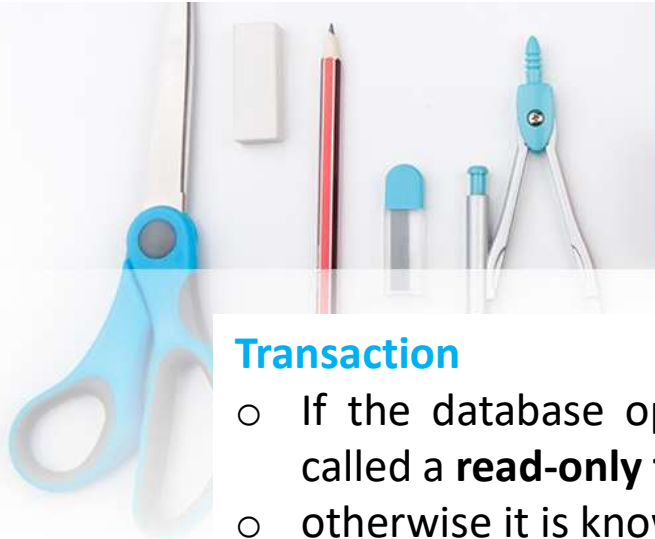
- **Multiuser:**
- It is multiuser if many users can use the system and hence access the database concurrently.



Introduction to Transaction Processing Concepts

Transaction

- Transaction is used to represent a logical unit of database processing that must be completed in its entirety to ensure correctness.
- A transaction is typically implemented by a computer program, which includes database commands such as:
 - **insertions,**
 - **retrievals,**
 - **deletions, and**
 - **updates.**
- The database operations that form a transaction can either be
 - a) **embedded within an application program or**
 - b) **they can be specified interactively via a high-level query language such as SQL.**



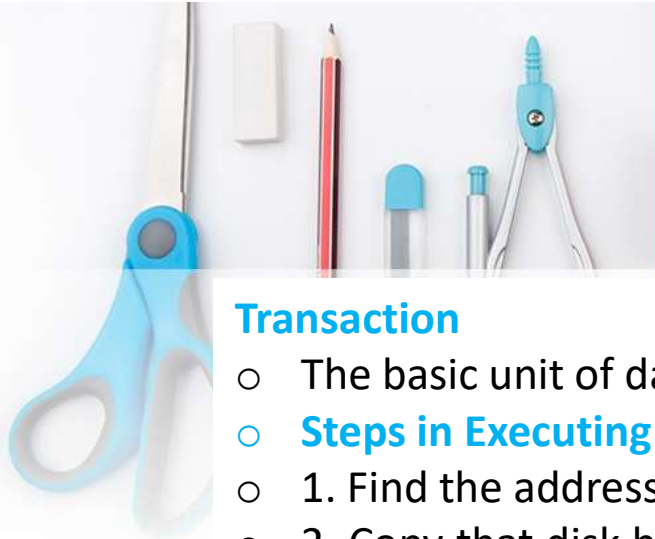
Introduction to Transaction Processing Concepts

Transaction

- If the database operations in a transaction only retrieve data, the transaction is called a **read-only transaction**;
- otherwise it is known as a **read-write transaction**.

Database access operations

- Basic database access operations that a transaction can include are as follows:
- **read_item(X)**. Reads a database item named X into a program variable. To simplify our notation, we assume that the program variable is also named X.
- **write_item(X)**. Writes the value of program variable X into the database item named X.



Introduction to Transaction Processing Concepts

Transaction

- The basic unit of data transfer from disk to main memory is one **disk block**.
- **Steps in Executing a read_item(X) command :**
 - 1. Find the address of the disk block that contains item X.
 - 2. Copy that disk block into a buffer in main memory (if that disk block is not already in some main memory buffer).
 - 3. Copy item X from the buffer to the program variable named X.
- **Steps in Executing a write_item(X) command :**
 - 1. Find the address of the disk block that contains item X.
 - 2. Copy that disk block into a buffer in main memory (if that disk block is not already in some main memory buffer).
 - 3. Copy item X from the program variable named X into its correct location in the buffer.
 - 4. Store the updated block from the buffer back to disk (either immediately or at some later point in time).



Concurrency Control

Concurrency Control

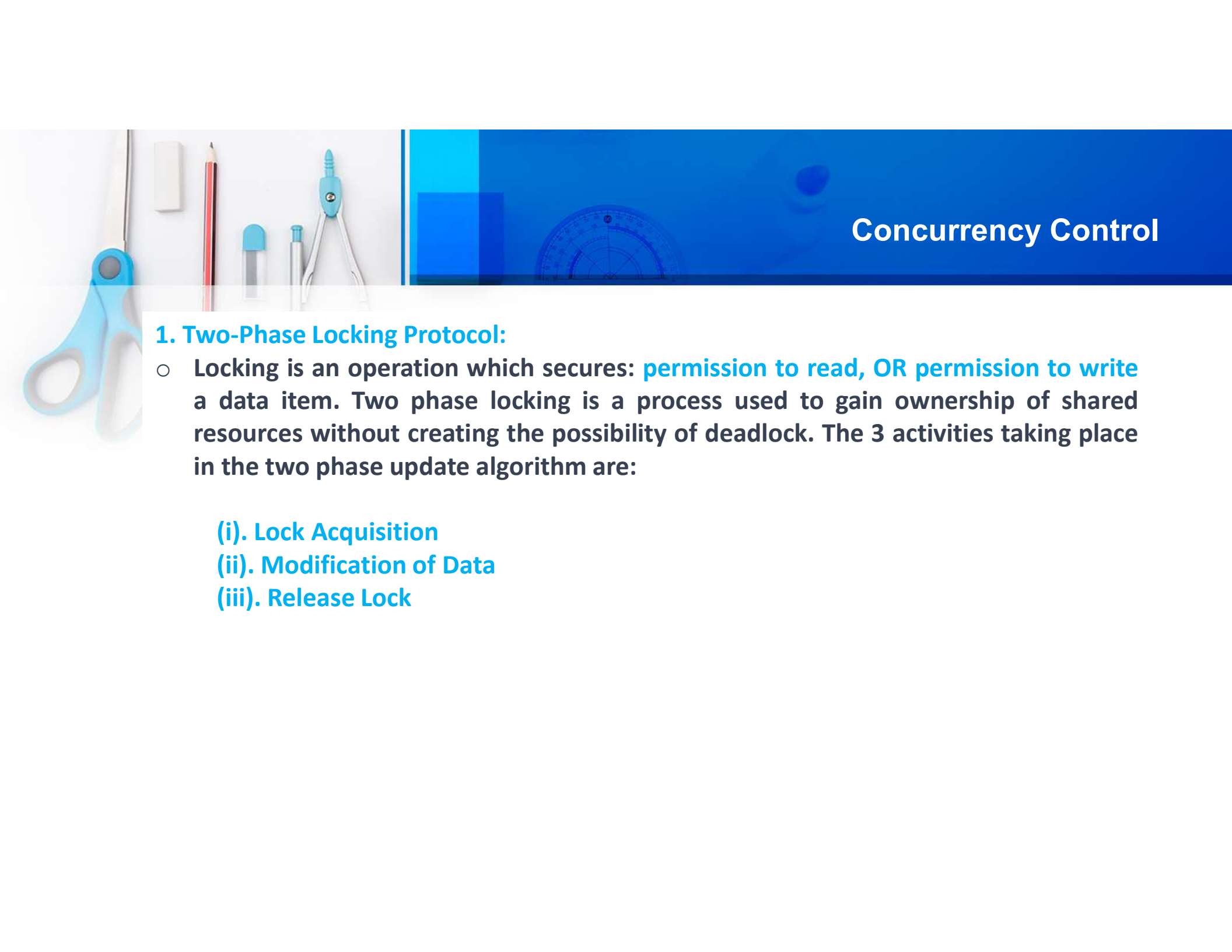
- Concurrent control is a very important concept of DBMS which ensures the simultaneous execution or manipulation of data by several processes or user without resulting in data inconsistency. Concurrency Control deals with interleaved execution of more than one transaction.
- Concurrency control problem, which occurs when multiple transactions submitted by various users interfere with one another in a way that produces incorrect results.



Concurrency Control

Techniques for concurrency control :

1. Two-phase locking Protocol
2. Time stamp ordering Protocol
3. Multi version concurrency control
4. Validation concurrency control



Concurrency Control

1. Two-Phase Locking Protocol:

- Locking is an operation which secures: **permission to read, OR permission to write** a data item. Two phase locking is a process used to gain ownership of shared resources without creating the possibility of deadlock. The 3 activities taking place in the two phase update algorithm are:

(i). Lock Acquisition

(ii). Modification of Data

(iii). Release Lock

2. Time Stamp Ordering Protocol:

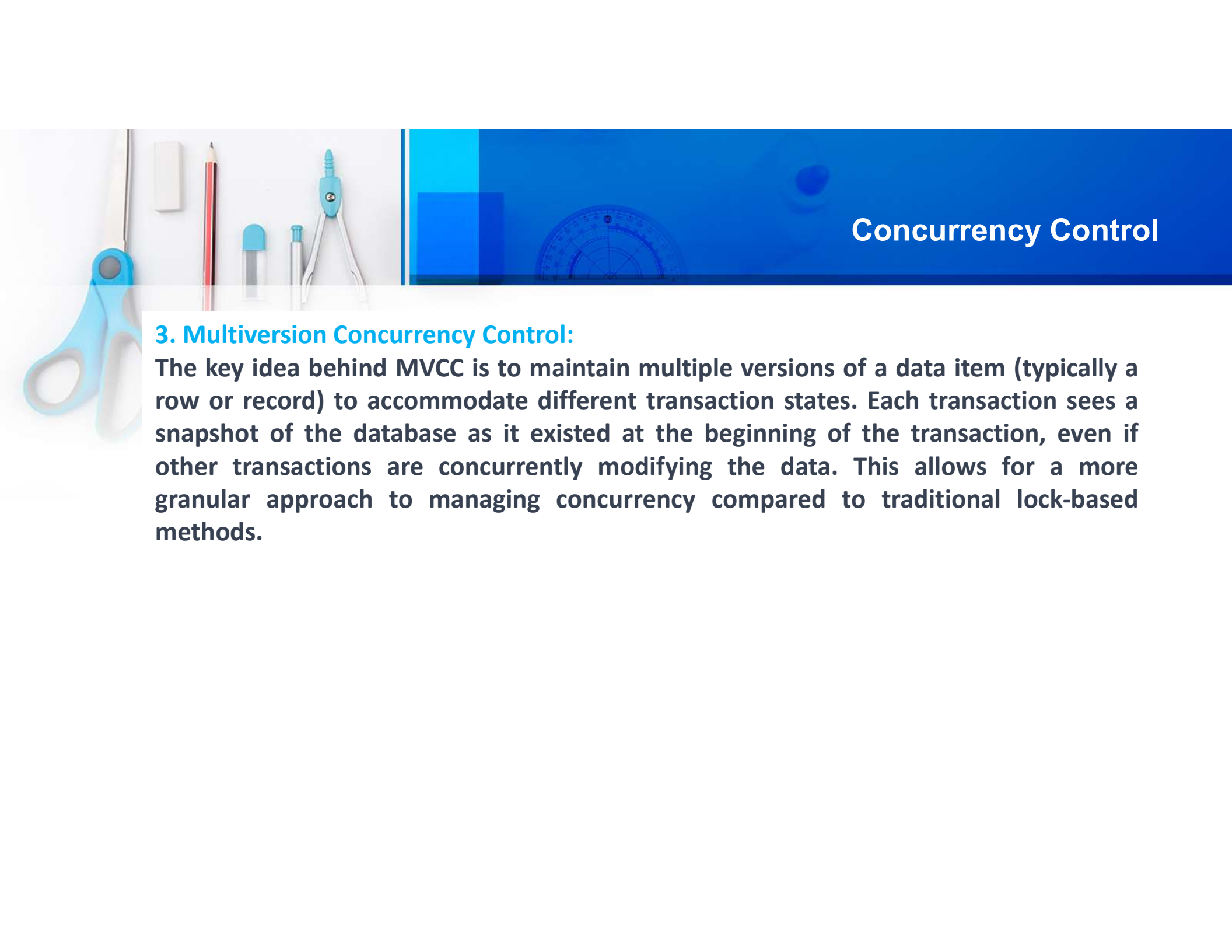
- A timestamp is a tag that can be attached to any transaction or any data item, which denotes a specific time on which the transaction or the data item had been used in any way. A timestamp can be implemented in 2 ways. One is to directly assign the current value of the clock to the transaction or data item. The other is to attach the value of a logical counter that keeps increment as new timestamps are required. The timestamp of a data item can be of 2 types:

(i) W-timestamp(X):

- This means the latest time when the data item X has been written into.

(ii) R-timestamp(X):

- This means the latest time when the data item X has been read from.
- These 2 timestamps are updated each time a successful read/write operation is performed on the data item X.



Concurrency Control

3. Multiversion Concurrency Control:

The key idea behind MVCC is to maintain multiple versions of a data item (typically a row or record) to accommodate different transaction states. Each transaction sees a snapshot of the database as it existed at the beginning of the transaction, even if other transactions are concurrently modifying the data. This allows for a more granular approach to managing concurrency compared to traditional lock-based methods.

4. Validation Concurrency Control:

- The optimistic approach is based on the assumption that the majority of the database operations do not conflict. The optimistic approach requires neither locking nor time stamping techniques. Instead, a transaction is executed without restrictions until it is **committed**. Using an optimistic approach, each transaction moves through 2 or 3 phases, referred to as read, validation and write.

(i) During read phase, the transaction reads the database, executes the needed computations and makes the updates to a private copy of the database values. All update operations of the transactions are recorded in a **temporary update** file, which is not accessed by the remaining transactions.

(ii) During the validation phase, the transaction is validated to ensure that the changes made will not affect the integrity and consistency of the database. If the validation test is positive, the transaction goes to a write phase. If the validation test is negative, the transaction is restarted and the changes are discarded.

(iii) During the write phase, the changes are permanently applied to the database.



Thank You all!